

Advanced Statistical Methods II

Lecture 13: Introduction to Robust Statistics

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Outline of Today's Lecture

After the mid-semester: a new theme

Motivation and historical background

Approximate parametric models

Fragility of classical estimators

Basic robust estimators

M-estimators

Measuring robustness

What comes next

Plan for the coming weeks



1. Robust Regression $\sim$ 2 weeks.
2. Non-linear Regression $\sim$ 3 weeks.
3. Ridge, Lasso and related topics $\sim$ 1 week.

methods where the **data structure** was non-standard (incomplete).

- Now we start another important theme:

A set of small navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other slide controls.

Today's main question



Classical statistical methods often begin with an ideal model:

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F_\theta.$$

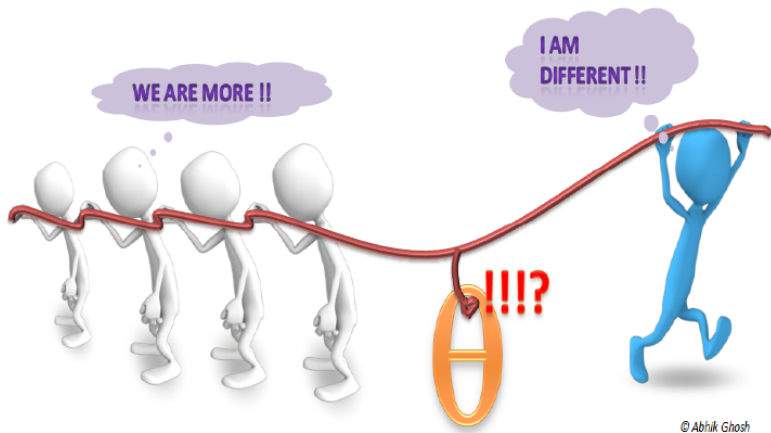
But in practice, we may have

most observations from F_θ , a few observations from something else.

Can an estimator remain useful when the assumed model is only approximately true?

This is the basic entry point into **robust statistics**.

Motivating picture



Source: lecture slides of [Basu \(2023\)](#).

Motivation and Background



- ▶ Often real data contain one or more “**outliers**”, observations that are incompatible with the majority of the data.
- ▶ The standard classical estimates, many of which have a high degree of efficiency at the true model, are often highly “**non-robust**” and easily influenced by outliers.

They include, among others, the **sample mean**, the **sample standard deviation** and the **least squares estimates** under normal linear regression.

Motivation and Background



- ▶ When we are trying to provide a **parametric fit** to a dataset and the data contain outliers, our aim is to come up with “**robust**” parameter estimates which provide good fit to the **majority of the data** by sacrificing the outliers. The robust parameter estimates should also provide a good fit to the entire data when there are no outliers.

Motivation and Background



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- ▶ The robust methods also help to provide useful outlier detection methods. After we model the data with a robust fit, the observations that stand out are the outliers. Such **outlier detection** is not possible with **non-robust fits**. This is particularly useful with multivariate data.

- For book-length treatments, see [Huber \(1981\)](#), [Rousseeuw and Leroy \(1987\)](#), and [Maronna et al. \(2006\)](#).

A Real Data Example: Drosophila Dataset

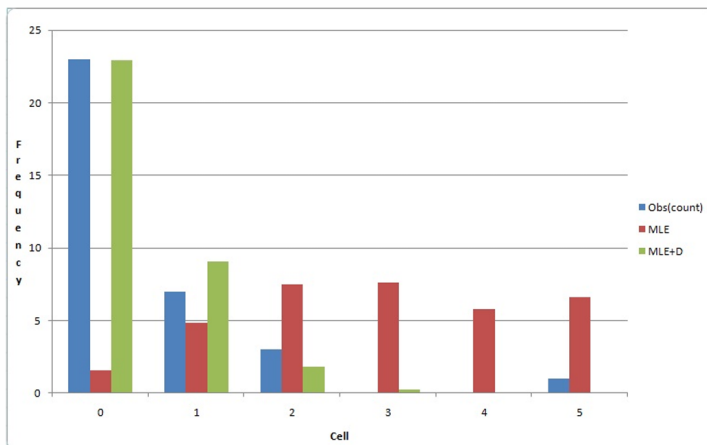


The experimenter exposed groups of male flies to different doses of a chemical to be screened. Subsequently each male was mated with unexposed females. Approximately 100 daughter flies were sampled for each male and the experimenter noted the number of daughters carrying a recessive lethal mutation on the X chromosome. The data set consisted of the observed frequencies of males having 0, 1, 2, ... recessive lethal daughters. Data for the specific experimental run on day 177 are presented as below.

Table 1: Drosophila data

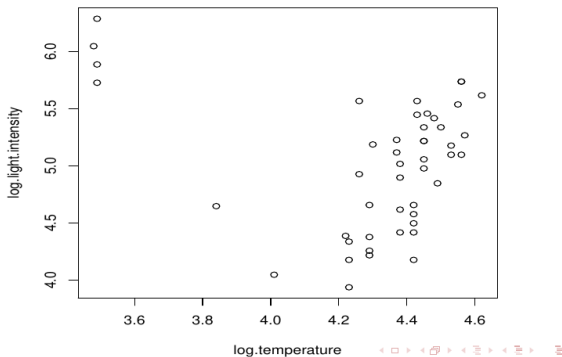
No. of daughters	0	1	2	3	4	≥ 5
Observed frequency	23	7	3	0	0	1(91)

Drosophila Data Example

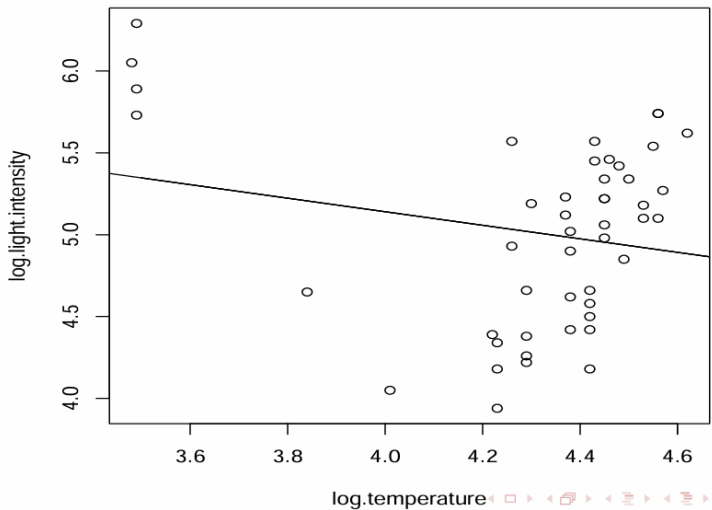


A regression example: star cluster data

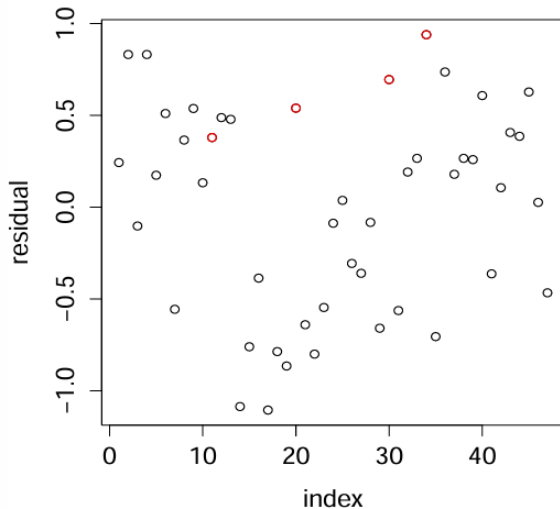
Figure: Hertzsprung-Russell Star Cluster Data



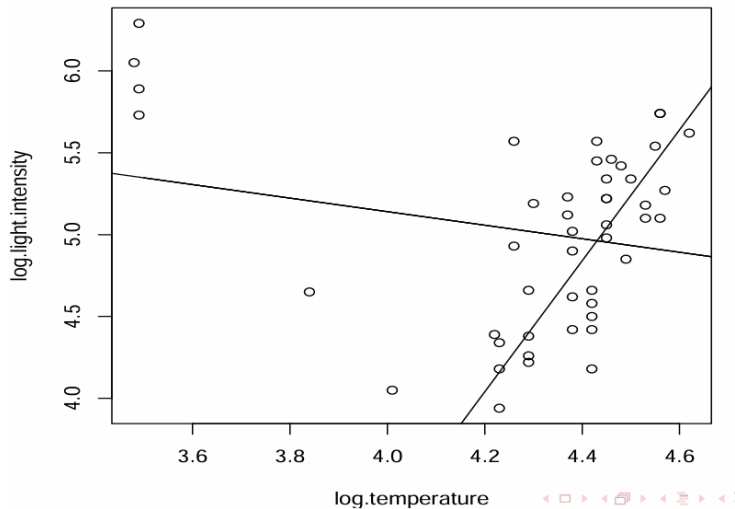
Star cluster data: LS fit



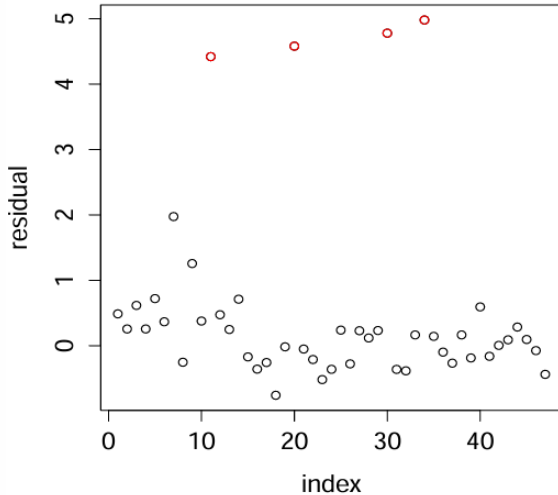
Star cluster data: Residuals after LS fit



Star cluster data: LS and LMS fit



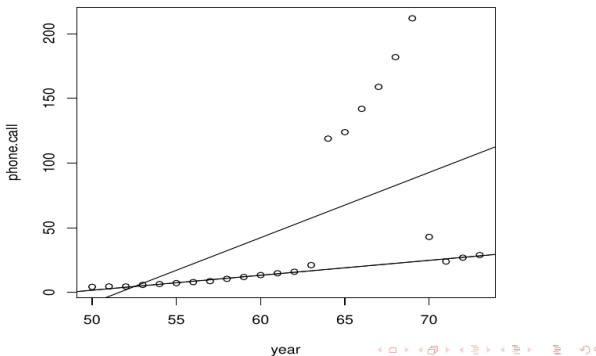
Star cluster data: Residuals after LMS fit



Another regression example: telephone data

Number of international phone calls (in tens of millions) from Belgium between 1950 and 1973.

Figure: Belgian Telephone Data (LS and LMS fit)



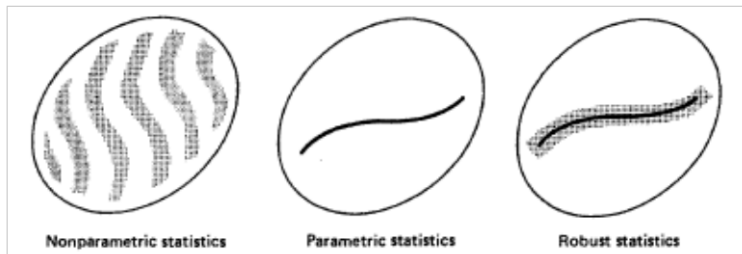
Parametric or Nonparametric?



Robust Statistics is the statistics of approximate parametric models.

Parametric, nonparametric, robust

Figure: Parametric versus Nonparametric



Source: [Hampel et al. \(1986\)](#).

A simple contamination model



A useful idealization is the **contamination neighborhood**

$$G_\varepsilon = (1 - \varepsilon)F_\theta + \varepsilon H, \quad 0 \leq \varepsilon \leq 1.$$

- ▶ F_θ : the main parametric model.
- ▶ H : an arbitrary contaminating distribution.
- ▶ ε : the contamination proportion.

If ε is small, we want an estimator that still behaves as if it is learning the main model F_θ .

This idea is connected to the contaminated-distribution framework discussed by [Tukey \(1960\)](#) and later formalized in robust theory.

Why approximate models are natural



A parametric model is usually a simplified mathematical description.

- ▶ Normality may be a good approximation for the central bulk, but not for the tails.
- ▶ A linear regression relation may describe most observations, but a few points may have high leverage.
- ▶ A Poisson model may fit ordinary counts, but one exceptional count can dominate the MLE.

Underlying difference

Classical inference asks: *What happens under the model?*

Robust inference asks: *What happens near the model?*

The sample mean can be fragile



Consider the ten observations

9, 10, 11, 10, 12, 9, 10, 11, 10, 100.

The sample mean can be fragile



Consider the ten observations

9, 10, 11, 10, 12, 9, 10, 11, 10, 100.

Estimator	Value	Comment
Mean	19.20	pulled strongly by 100
Median	10.00	stable central value
10% trimmed mean	10.375	discards the two extremes
Sample SD	28.41	inflated by the outlier
MAD/0.6745	1.48	robust scale estimate

One observation can move the mean far away from the typical observation.

For one-dimensional location, common robust alternatives to the mean include:

1. **Median**: middle observation after sorting.
2. **Trimmed mean**: remove a fixed fraction from each tail and average the rest.
3. **Winsorized mean**: replace tail observations by the most extreme retained observations.

All three reduce the ability of extreme observations to dominate the estimate.

Trimmed and Winsorized means



Suppose $x_{(1)} \leq \cdots \leq x_{(n)}$ are ordered observations.

For trimming proportion α , let $m = \lfloor n\alpha \rfloor$.

$$\bar{x}_{\text{trim}} = \frac{1}{n - 2m} \sum_{i=m+1}^{n-m} x_{(i)}.$$

The Winsorized sample replaces

$$x_{(1)}, \dots, x_{(m)} \quad \text{by} \quad x_{(m+1)},$$

and

$$x_{(n-m+1)}, \dots, x_{(n)} \quad \text{by} \quad x_{(n-m)}.$$

Then we take the ordinary mean of the modified sample.

Robust scale: median absolute deviation



The usual sample standard deviation can be heavily affected by outliers.

A common robust alternative is the median absolute deviation:

$$\text{MAD} = \text{median}_i |X_i - \text{median}_j(X_j)|.$$

For normal calibration, one often uses

$$\hat{\sigma}_{\text{MAD}} = \frac{\text{MAD}}{0.6745}.$$

The constant 0.6745 is approximately the median of $|Z|$ when $Z \sim N(0, 1)$.

Why least squares is vulnerable



In ordinary least squares, we minimize

$$\sum_{i=1}^n r_i^2, \quad r_i = y_i - x_i^\top \beta.$$

A large residual receives a **quadratically large** penalty.

Consequence

A single point with very large residual or very high leverage can have a large effect on the fitted regression line.

This motivates robust alternatives such as:

- ▶ least absolute deviations regression;
- ▶ Huber-type regression;
- ▶ least median of squares;
- ▶ robust M-estimation.

The M-estimation framework was developed in the foundations of

End of class – see M. estimate & printing

Location M-estimators

For a location parameter θ , a common form is

$$\sum_{i=1}^n \psi \left(\frac{x_i - \theta}{s} \right) = 0,$$

where s is a scale estimate.

Equivalently, many M-estimators minimize

$$\sum_{i=1}^n \rho \left(\frac{x_i - \theta}{s} \right), \quad \psi = \rho'.$$

Loss $\rho(u)$	Score $\psi(u)$	Behavior
$u^2/2$	u	mean/MLE under normality
$ u $	$\text{sign}(u)$	median/LAD style
Huber loss	clipped linear score	compromise

Huber's score function



Huber's score function is

$$\psi_c(u) = \begin{cases} u, & |u| \leq c, \\ c \operatorname{sign}(u), & |u| > c. \end{cases}$$

- ▶ Small residuals are treated approximately as in least squares.
- ▶ Large residuals receive bounded influence through clipping.
- ▶ The tuning constant c controls the robustness-efficiency trade-off.

$$T(F),$$

- ▶ Mean functional: $T(F) = \int x \, dF(x)$.
- ▶ Median functional: $T(F) = F^{-1}(1/2)$.
- ▶ M-estimator functional: solution of $\int \psi_\theta(x) \, dF(x) = 0$.

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where Δ_y puts all mass at the point y .

$$F_\varepsilon = (1 - \varepsilon)F + \varepsilon\Delta_v,$$

$$\text{IF}(y; T, F) = \left. \frac{\partial}{\partial \varepsilon} T(F_\varepsilon) \right|_{\varepsilon=0}.$$

Influence function: intuition



Consider the contaminated distribution

$$F_\varepsilon = (1 - \varepsilon)F + \varepsilon\Delta_y,$$

where Δ_y puts all mass at the point y .

The influence function is

$$\text{IF}(y; T, F) = \left. \frac{\partial}{\partial \varepsilon} T(F_\varepsilon) \right|_{\varepsilon=0}.$$

It measures the infinitesimal effect of putting a tiny contamination at y .

Influence function for the mean



For the mean functional,

$$T(F) = \int x \, dF(x).$$

Then

$$T(F_\varepsilon) = (1 - \varepsilon)T(F) + \varepsilon y.$$

Therefore,

$$\text{IF}(y; T, F) = y - T(F).$$

This is unbounded in y . Hence the sample mean is non-robust against arbitrarily large contamination.

$$\text{IF}(y; T, F) = \frac{1}{2f(T(F))} \text{sign}(y - T(F)),$$

For a formal treatment, see [Hampel \(1971\)](#) and [Hampel et al. \(1986\)](#).

Breakdown point: another robustness measure



The finite-sample breakdown point asks:

What fraction of observations must be replaced by arbitrary values before the estimator can be driven arbitrarily far away?

Breakdown point: another robustness measure



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What fraction of observations must be replaced by arbitrary values before the estimator can be driven arbitrarily far away?

Estimator	breakdown	Interpretation
Mean	about $1/n$	one huge value can destroy it
Median	about 50%	needs nearly half the data to be bad
Least squares	small	sensitive to leverage/outliers
LMS	high	designed for outlying regression data

This idea is central in high-breakdown robust regression; see [Rousseeuw and Leroy \(1987\)](#).

► **efficiency** under the exact assumed model:

- **robustness** under contamination or model misspecification:

if a small fraction of observations is contaminated, the estimate should not change wildly.

► **efficiency** under the exact assumed model:

- **robustness** under contamination or model misspecification:

But the trade-off is not always severe.

In later lectures, this theme will appear in robust regression, LAD regression, and quantile regression.

Today's concepts in one table



Concept	Main idea
Outlier	observation incompatible with the majority pattern
Contamination model	small fraction may come from another distribution
Robust estimator	stable under small model deviations
Median/trimmed mean	reduce effect of extreme observations
MAD	robust measure of scale
M-estimator	solve modified estimating equation
Influence function	infinitesimal effect of contamination at a point
Breakdown point	fraction of contamination needed to destroy estimator

Key takeaway messages



1. Robust statistics studies inference under approximate parametric models.
2. Classical estimators such as the mean, standard deviation, MLE, and least squares can be highly sensitive to outliers.
3. Robust estimators deliberately reduce the influence of extreme observations.
4. M-estimation provides a bridge between maximum likelihood and robust alternatives.
5. Influence functions and breakdown points help quantify robustness.

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